THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE FULL SYLLABUS PAPER SET 08

Full Syllabus (Chapters 1–13) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ is a: 1

(a) combination reaction **(b)** decomposition reaction
(c) displacement reaction **(d)** double displacement reaction

2. The indicator that is red in acid and yellow in base is: 1

(a) turmeric **(b)** methyl orange **(c)** red litmus **(d)** phenolphthalein

3. Blue copper sulphate crystals turn white on heating because they lose: 1

(a) sulphur **(b)** oxygen
(c) copper **(d)** water of crystallisation

4. The metal at the top of the reactivity series (among these) is: 1

(a) potassium **(b)** copper **(c)** zinc **(d)** gold

5. Highly reactive metals are extracted by: 1

(a) reduction with carbon **(b)** heating alone
(c) roasting only **(d)** electrolytic reduction

6. The structural isomers possible for C_4H_{10} are: 1

(a) 3 **(b)** 4 **(c)** 1 **(d)** 2

7. Ethanol on dehydration gives: 1

(a) ethene **(b)** ethanal **(c)** ethyne **(d)** ethanoic acid

8. For blood pressure, Student A says systolic pressure is ~120 mm Hg; Student B says diastolic is ~80 mm Hg. Then: 1

(a) Both are correct **(b)** Only Student A is correct
(c) Only Student B is correct **(d)** Both are wrong

9. The growth of roots towards water is called: 1

(a) chemotropism **(b)** geotropism **(c)** hydrotropism **(d)** phototropism

10. A student states: “Hydra reproduces by budding as well as regeneration.” The correct response is: 1

(a) Correct **(b)** Incorrect
(c) Only budding **(d)** Cannot be determined

11. The movement of pollen from one flower to another of the same species is: 1

(a) fertilisation **(b)** self-pollination
(c) cross-pollination **(d)** germination

12. In a dihybrid cross, the number of phenotypes in F2 is: 1

(a) 9 **(b)** 4 **(c)** 2 **(d)** 3

13. The number of autosomes in a human somatic cell is: 1

(a) 44 **(b)** 23 **(c)** 46 **(d)** 22

14. Two lenses of power +3 D and −1 D are in contact. The net power is: 1

(a) −2 D **(b)** +2 D **(c)** +4 D **(d)** +3 D

15. Which part of the eye has cells sensitive to colour? 1

(a) cones in the retina **(b)** rods in the cornea
(c) lens **(d)** iris

16. For a fuse wire, Student A says it should have a low melting point; Student B says it should have a high melting point. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

17. Power consumed by a 10 Ω resistor carrying 2 A is: 1

(a) 20 W **(b)** 5 W **(c)** 200 W **(d)** 40 W

18. Which instrument detects a magnetic field? 1

(a) compass **(b)** voltmeter **(c)** rheostat **(d)** ammeter

19. Which of these will be broken down by the action of bacteria? 1

(a) glass bottle **(b)** polythene bag **(c)** aluminium foil **(d)** cotton cloth

20. An organism which feeds on both plants and animals is a/an: 1

(a) carnivore **(b)** omnivore **(c)** herbivore **(d)** producer

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. A student says oily food becomes rancid due to reduction. Justify or refute. 2

22. A student says that acids and bases neutralise each other only in the laboratory. Justify or refute. 2

23. Name the two types of cells in which anaerobic respiration occurs. 2

24. How does the shoot respond to gravity? 2

25. Name the parts of the pistil. 2

26. What is the F1 phenotype when pure tall (TT) is crossed with pure short (tt)? 2

27. A student says that the refractive index has a unit. Justify or refute. 2

28. What does the crowding of field lines indicate? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Distinguish between displacement and double displacement reactions with one example each. 3

30. What happens when a metal reacts with an acid? Write the equation of magnesium with dilute sulphuric acid. Name the gas and give its test. 3

31. What is the difference between a reflex action and walking? 3

32. What are the disadvantages of surgical methods of contraception? 3

33. Explain the Tyndall effect. Where can it be observed? 3

34. A 15 A circuit is used for heaters while a 5 A circuit is used for bulbs. Explain why. 3

35. Write two uses each of bleaching powder and baking soda. 3

OR

Name the gas evolved and write the equation when 3
(i) zinc reacts with dilute sulphuric acid
(ii) sodium hydrogencarbonate reacts with dilute HCl
(iii) zinc reacts with sodium hydroxide.

36. How is ozone formed? Why is its depletion a cause of concern? Name the chemical responsible. 3

OR

What is the function of ozone in the upper atmosphere? How is it formed? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) What is refraction? State its laws. 5
(ii) What is the absolute refractive index?
(iii) Explain why a coin in a bucket of water appears raised.

OR

A 4 cm tall object is placed 15 cm from a convex lens of focal length 10 cm. 5
(i) Find the position, size and nature of the image.
(ii) Draw the ray diagram (describe).
(iii) What is the magnification?

38. **Case study:** Hydrocarbons P (C_3H_8), Q (C_3H_6) and R (C_3H_4) are tested. 5
(i) Classify each as alkane/alkene/alkyne.
(ii) Which ones decolourise bromine water?
(iii) Write the structure of Q.
(iv) Which one burns with the cleanest flame?

OR

Case study: Compound X (C_2H_6O) is used as an antiseptic and in cough syrups. On heating with conc. H_2SO_4 at 443 K it gives Y, which decolourises bromine water. 5
(i) Identify X and Y.
(ii) Write the equation.
(iii) What is the role of conc. H_2SO_4?
(iv) Write the reaction of X with sodium.

39. (i) Derive the expression for equivalent resistance of three resistors in series. 5

(ii) Two resistors of 5 Ω and 10 Ω are connected in series to a 6 V battery. Find the current and the potential difference across each.

OR

Case study: A house has 4 bulbs of 60 W each used 5 h daily and a 1000 W heater used 2 h daily at 220 V. 5

(i) Find the total energy consumed per day in kWh.

(ii) Find the monthly (30 days) cost at ₹5 per unit.

(iii) Find the current drawn by the heater.

(iv) Why is a fuse used?

40. (i) Describe the alimentary canal of man (main parts). 5

(ii) What is peristalsis?

(iii) Function of the large intestine?

OR

Case study: Destarched potted plants were used. One leaf was covered with black paper strips, and the plant was kept in sunlight for 6 h. The leaf was then tested with iodine. 5

(i) Why was the plant destarched?

(ii) What colour do covered and uncovered parts give?

(iii) What conclusion is drawn?

(iv) Why is the leaf boiled in alcohol?

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND